

SMS-based payments are extensively used for airline ticket booking, purchase of online games, music CDs, movies and videos, etc. Prominent players in this area are PayMate, mCheck and Gpay.

**Direct mobile billing.** In this mode of mobile payment, the customer uses his mobile phone for purchase of digitised items such as mobile ringtones, music, videos, video games and wallpapers that can be downloaded in the mobile device itself. The purchase amount is added to the monthly mobile bill of the customer.

This provides an alternative payment method that does not require use of credit/debit cards or any online payment solution provider, such as PayPal, and thus bypasses banks and credit card companies altogether. The customer is billed by the mobile network operator based on different tariff rates, and the payment is either debited

payments, the mobile user initiates the transaction by entering the PIN from his mobile handset. The issuing bank verifies the PIN and authorises the payment. Next, the customer enters the amount to be paid and completes the transaction. The amount is automatically deducted from the credit card account of the mobile user and credited to the bank account of the payee business partner such as the shop owner.

In order to ensure safe and secure payments, the issuing bank employs user authentication using encryption technique. In case the mobile handset is stolen or lost, the corresponding PIN is automatically and instantly locked so that no payment can be made using the mobile handset. As the customer only needs to enter the PIN and no other details such as credit card number, date of birth and date of expiry are required, such mobile credit card based payments provide a much simpler and

sonal data such as credit card number, driving licence number, bank account number, insurance information and personal identification number (PIN). These are primarily used for user identification, authentication and data storage.

Smart cards are of two types, contact and contactless. Contact smart cards need a card reader to retrieve the stored data and transmit to external devices such as computers, point-of-sale (POS) terminals or mobile phones for further processing. These have a contact area containing some gold-plated contact pads. When the contact smart card is swiped inside the card reader, contact pads in the smart card make contact with electrical connectors of the card reader and allow reading the information from the microprocessor chip or writing the information back to the memory of the smart card.

In contactless smart cards, the microprocessor chip communicates with the card reader through radio frequency identification (RFID) — a technology that employs radio waves to transmit/receive information to/from the card reader without any physical contact. It requires only an antenna in close proximity to complete the transmission. This provides a hands-free, contact-free way of transmitting data from the smart card to external devices such as POS terminals or fare collection terminals in mass transit systems.

Contactless smart cards are extensively used in mobile payment systems. Mobile phones come with a subscriber identity module (SIM), which is a microprocessor-based card containing programmable memory to store user information, used for authentication purposes. If the smart card capabilities are combined with SIM cards, the mobile phone can very well act as a contactless smart card and can be used effectively in making mobile payments. Both the smart card and SIM card are integrated in a mobile handset and the mobile phone is used for making payments in a contactless mode.

## Mobile phones equipped with contactless smart cards use near-field communication technology to exchange data between the mobile device and nearby smart card readers

from the subscriber's prepaid account or added to the standard postpaid invoice of the subscriber as the case may be. In order to secure the payment, two-stage authentication involving PIN and one-time-password is used while making the purchase through the mobile device.

In most of the cases, the transaction completes in less than 10 seconds. As this provides a fast and easy way of mobile purchase, around 70 per cent of all digital content purchased online uses direct mobile billing method.

### Card-based mobile payments

**Credit card based payments.** In credit card based mobile payment system, the mobile handset is used as a credit card for making payments. The credit card issuing bank gives a PIN number to the mobile handset user. In order to make

convenient mode of mobile payment.

Mobile handset users are using this mode of payment extensively for post-paid bill payment, booking movie tickets and travel tickets, micro payments in shops and restaurants, and shopping on the Internet. Leading players in this field are Paymo in USA, DoCoMo in Japan and Reliance mPay in India. Online payment companies such as PayPal, Amazon Payment and Google Checkout also provide mobile options. These require the customers to register with a PIN that they will use for making mobile payments from their mobile handsets. These systems are integrated with credit card payment systems through a combined Web payment interface.

**Smart card based payments.** Smart cards are plastic cards with embedded integrated circuits containing microprocessors and memory to store per-